

# Md. Abdur Rahman Shibly

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Dhaka, Bangladesh

## Education

Bangladesh University of Engineering and Technology (BUET)

March 2018 – May 2023

Bachelor of Science

CGPA: 3.29 / 4.00

BSc Thesis:

Building a Semantic Search Engine using an Ontology: A Case Study on Social Media Influencers.

## Research Interests

- Applied AI across interdisciplinary domains
- NLP
- RAG
- HCI
- Conversational AI
- Software Engineering
- Hallucination Reduction in AI Agents

## Ongoing Research

Semantic Search Engine using an Ontology

January 2023 - Current

Tools: Protege, Neo4j, Django, Python, NumPy, Pandas

[Supervisor]

Developed a novel ontology for social media influencers to categorize them by audience location, interaction rates, and content types. Current work focuses on refining assumptions from prior research by fine-tuning LLaMA2 and utilizing prompt engineering techniques.

Online Facility Assignment with Nonuniform Capacities

April 2025 - Current

Tools: Python, Pandas, NumPy, SciPy, Matplotlib

[Supervisor]

Investigating the online facility assignment problem with nonuniform facility capacities in metric spaces. Established a reduction from the general setting to the unit-capacity case, simplifying the analysis of greedy algorithms. Derived tighter competitive ratio bounds for algorithms on Euclidean planes and unweighted graphs, extending prior results on uniform capacities.

## Publications

### Conference Papers

- Shibly, M. A. R., et al., "Leveraging the Domain Adaptation of Retrieval-Augmented Generation Models for Question Answering," in *Proceedings of the 22nd International Conference on Information Technology – New Generations (ITNG 2025)*, Springer, 2025. DOI

### Book Chapters

- Chapter in Springer Nature book collection based on ITNG 2025 conference proceedings. [Link](#)

## Test Scores

IELTS (August 2025): Overall Band **8.0** (Reading: 9, Listening: 8, Speaking: 8, Writing: 6.5)

## Professional Experience

Software Engineer

March 2025 – Present

Bengal Mobile QA Solutions

Dhaka, Bangladesh

- Developed AI-powered systems, including AI Ruler, a dynamic parser for structuring raw CDRs
- Built chatbots and personalized AI assistants using LLMs for hotel interactions and parent-focused applications
- Led development of Invoice-Factory-AI, applying prompt engineering to extract structured data from invoices
- Modernized hotel management system with microservices-based architecture for scalability and maintainability

Assistant Software Engineer

July 2024 – February 2025

Computer Network Systems

Dhaka, Bangladesh

- Co-led government insurance project BISDP-BI, overseeing data analysis and ML model development for intelligent decision-making and automation
- Collaborated on AI-driven projects including Aptitudo and Zeuxis.AI, contributing to model development and data analysis
- Worked with cross-functional teams to design, test, and integrate AI/ML components into production-level solutions

Consultant

December 2023 – July 2024

Bengal Mobile QA Solutions

Dhaka, Bangladesh

- Developed AI assistant for hotel industry using LLMs, focusing on conversational AI solutions tailored for hotel-specific tasks
- Applied expertise in LLM-based systems for creating specialized industry applications

Projects

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**BISDP-BI**  
*Government Insurance Project*

August 2023 – August 2024

Developed comprehensive government insurance project focusing on data analysis and ML model development for intelligent decision-making regarding fraudulent claims and monetary predictions. Co-led the machine learning team throughout the entire project lifecycle.

**Python, Pandas, NumPy, Scikit-learn, Machine Learning Models**

**Invoice-Factory-AI**  
*AI-Powered Invoice Processing*

May 2025 – July 2025

Developed AI system using LLM and prompt engineering to parse hotel career invoices of various formats, extracting structured data automatically. Integrated Google Document AI for OCR capabilities.

**Django, LLM, Prompt Engineering, Google Document AI**

**Parental Assistant**  
*Multi-Agent AI System*

August 2025 – Present

Building comprehensive AI assistant for parents to help monitor and support their children using advanced multi-agent system architecture.

**Langgraph, Langchain, Langfuse, Multi-agent Systems**

**Zeuxis.AI**  
*Multimedia Generation Platform*

July 2024 – December 2024

Developed comprehensive platform enabling users to generate images and videos from text or image inputs, including face swapping capabilities in a unified interface.

**Java, Spring Boot, FAL AI, Vision Models**

**Aptitudo**  
*AI Interview Assistant*

May 2024 – August 2024

Created AI-powered interview preparation system generating diverse question types including MCQ, coding, and written questions, with automated scoring capabilities.

**Django, LLM, Prompt Engineering**

References

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**Dr. Muhammad Masroor Ali**  
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Bangladesh University of Engineering and Technology  
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Relationship: Research Supervisor